

THE CHI-SQUARE DISTRIBUTION

In 1900, Karl Pearson showed that “as n approaches infinity, a discrete multinomial distribution approach to a Chi-square distribution”. **It is to be noted that the Chi-Square distribution have only one parameter which is its degree of freedom.**

DEGREE OF FREEDOM (d.f.):

Degree of Freedom is defined as “the number of independent values that a statistical analysis can estimate.”

The Chi-Square (χ^2) Distribution:

Let $Z_1, Z_2, \dots, Z_n$ be normally and independently distributed variables with zero means and unit variances. Then a random variable

$$\chi^2 = \sum Z_i^2$$

is defined as a Chi-square random variable with n degrees of freedom. The *p.d.f.* of χ^2 distribution is;

$$f(\chi^2) = \frac{1}{2^{n/2} \Gamma(n/2)} (\chi^2)^{\frac{n}{2}-1} \cdot e^{-\left(\frac{\chi^2}{2}\right)}; \quad 0 \leq \chi^2 \leq \infty$$

SOME PROPERTIES OF CHI-SQUARE DISTRIBUTION:

- (i) The Chi-square is a continuous distribution ranging from *zero* to *infinity*
- (ii) The mean of a Chi-square distribution is equal to the number of degrees of freedom and its variance is equal to the twice the number of degrees of freedom.

$$i.e. \quad E(\chi^2) = n, \quad Var(\chi^2) = 2n$$

- (iii) The χ^2 distribution tends to normal distribution as the number of degrees of freedom approaches to infinity.

TESTING HYPOTHESIS ABOUT p's OF THE MULTINOMIAL DISTRIBUTION:

- (i) H_0 : Data Agree with theory
 H_1 : Data do not Agree with theory
- (ii) **Level of Significance:**
- (iii) **Test Statistic:**

$$\chi^2 = \sum \left[\frac{(f_o - f_e)^2}{f_e} \right]$$

Where f_o = Observed frequency; f_e = Expected or theoretical frequency

- (iv) **Calculation:**

(v) Critical region:

$$\text{If } \chi^2 \geq \chi^2_{(\alpha, d.f.)} \quad d.f. = k - 1$$

Then we shall reject H_0

(vi) Conclusion:

Example# 1: In a genetic experiment involving the crossing of two types of peas, Mendal observed 315 round and yellow, 108 round and green, 101 angular and yellow, and 32 angular and green seeds from the plants but his theory of heredity called for a 9 : 3 : 3 : 1 ratio. Using a 5% level of significance, do the data support the theory?

Solution:

- (i) H_0 : Data Agree with theory
 H_1 : Data do not Agree with theory

(ii) Level of Significance:

$$\alpha = 0.05$$

(iii) Test Statistic:

$$\chi^2 = \sum \left[\frac{(f_0 - f_e)^2}{f_e} \right]$$

(iv) Calculation:

Given ratio is 9 : 3 : 3 : 1

Categories	f_0	p_i	$f_e = np_i$	$(f_0 - f_e)^2$	$\frac{(f_0 - f_e)^2}{f_e}$
Round & Yellow	315	9/16	$556 \times \frac{9}{16} = 312.75$	5.0625	0.016
Round & green	108	3/16	$556 \times \frac{3}{16} = 104.25$	14.0625	0.135
Angular & Yellow	101	3/16	$556 \times \frac{3}{16} = 104.25$	10.5625	0.101
Angular & green	32	1/16	$556 \times \frac{1}{16} = 34.75$	7.5625	0.218
Total	$\sum f_0 = n = 556$	1			$\chi^2 = 0.470$

$$\chi^2 = \sum \left[\frac{(f_0 - f_e)^2}{f_e} \right] = 0.470$$

(v) Critical region:

$$\text{If } \chi^2 \geq \chi^2_{(\alpha, d.f.)} = \chi^2_{(0.05, 3)} = 7.82$$

$$d.f. = k - 1 = 4 - 1 = 3$$

Then we shall reject H_0

(vi) Conclusion:

Since the calculated value does not lie in the critical region, therefore the null hypothesis cannot be rejected. It is therefore concluded that data support the theory.

Example# 2: A machine is supposed to mix peanuts, hazelnuts, cashews and pecans in the ratio 5 : 2 : 2 : 1. A can containing 500 of these mixed nuts was found to have 269 peanuts, 112 hazelnuts, 74 cashews and 45 pecans. At the 0.05 level of significance, test the hypothesis that the machine is mixing the nuts in the supposed ratio.

Solution:

- (i) H_0 : Data Agree with theory
 H_1 : Data do not Agree with theory
- (ii) **Level of Significance:**
 $\alpha = 0.05$
- (iii) **Test Statistic:**

$$\chi^2 = \sum \left[\frac{(f_0 - f_e)^2}{f_e} \right]$$

(iv) **Calculation:**

Given ratio is 5 : 2 : 2 : 1

Categories	f_0	p_i	$f_e = np_i$	$(f_0 - f_e)^2$	$\frac{(f_0 - f_e)^2}{f_e}$
Peanuts	269	5/10	$500 \times \frac{5}{10} = 250$	361	1.444
Hazelnuts	112	2/10	$500 \times \frac{2}{10} = 100$	144	1.44
Cashews	74	2/10	$500 \times \frac{2}{10} = 100$	676	6.76
Pecans	45	1/10	$500 \times \frac{1}{10} = 50$	25	0.5
Total	$\sum f_0 = n = 500$	1			$\chi^2 = 10.144$

$$\chi^2 = \sum \left[\frac{(f_0 - f_e)^2}{f_e} \right] = 10.144$$

(v) **Critical region:**

If $\chi^2 \geq \chi^2_{(\alpha, d.f.)} = \chi^2_{(0.05, 3)} = 7.82$

$d.f. = k - 1 = 4 - 1 = 3$

Then we shall reject H_0

(vi) **Conclusion:**

Since the calculated value falls in the critical region, therefore the null hypothesis rejected. It is therefore concluded that the hypothesis that the machine is not mixing the nuts in the supposed ratio.

Example# 3: Each of 126 individuals of a certain mammal species was placed in an enclosed container equal amount of each of 6 different foods. The frequency with which the animal chose each of the food was as follows;

Food items	1	2	3	4	5	6	Total
Frequency	13	26	31	14	28	14	126

Test the hypothesis that is there any preference among the food item using χ^2 test at 5% level of significance.

Solution:

(i) H_0 : Data Agree with theory

H_1 : Data do not Agree with theory

(ii) **Level of Significance:**

$$\alpha = 0.05$$

(iii) **Test Statistic:**

$$\chi^2 = \sum \left[\frac{(f_0 - f_e)^2}{f_e} \right]$$

(iv) **Calculation:**

Food items	f_0	p_i	$f_e = np_i$	$(f_0 - f_e)^2$	$\frac{(f_0 - f_e)^2}{f_e}$
1	13	1/6	$126 \times \frac{1}{6} = 21$	64	3.05
2	26	1/6	$126 \times \frac{1}{6} = 21$	25	1.19
3	31	1/6	$126 \times \frac{1}{6} = 21$	100	4.76
4	14	1/6	$126 \times \frac{1}{6} = 21$	49	2.33
5	28	1/6	$126 \times \frac{1}{6} = 21$	49	2.33
6	14	1/6	$126 \times \frac{1}{6} = 21$	49	2.33
Total	$\sum f_0 = n = 126$	1			$\chi^2 = 15.99$

(v) **Critical region:**

$$\text{If } \chi^2 \geq \chi^2_{(\alpha, d.f.)} = \chi^2_{(0.05, 5)} = 11.07$$

$$d.f. = 6 - 1 = 6 - 1 = 5$$

Then we shall reject H_0

(vi) **Conclusion:** Since the calculated value falls in the critical region, therefore the null hypothesis rejected. It is therefore concluded that data do not support the theory.

Example# 4: The grades in a statistics course were as follows;

Grade	A	B	C	D	E
f	14	18	32	20	16

Test the hypothesis at the 0.05 level of significance, that the distribution of grades is uniform.

Solution:

- (i) (ii) (iii)
(iv) **Calculation:**

Grade	f_0	p_i	$f_e = np_i$	$(f_0 - f_e)^2$	$\frac{(f_0 - f_e)^2}{f_e}$
A	14	1/5			
B	18	1/5			
C	32	1/5			
D	20	1/5			
E	16	1/5			
Total	$\Sigma f_0 = n =$	1			$\chi^2 =$

(v) **Critical region:**

(vi) **Conclusion:** D.Y.S.

Example# 5: A random number table of 250 digits showed the following distribution of the digits 0, 1, 2,.....,9.

Digit	0	1	2	3	4	5	6	7	8	9
frequency	17	31	29	18	14	20	35	30	20	36

Test the hypothesis, at the 0.05 level of significance, that the digits were distributed in equal numbers in the table.

Solution:

- (i) (ii) (iii)
(iv) **Calculation:**

Digits	f_0	p_i	$f_e = np_i$	$(f_0 - f_e)^2$	$\frac{(f_0 - f_e)^2}{f_e}$
0	17	1/10			
1	31	1/10			
2	29	1/10			
3	18	1/10			
4	14	1/10			
5	20	1/10			
6	35	1/10			
7	30	1/10			
8	20	1/10			
9	36	1/10			
Total	$\Sigma f_0 = n =$	1			$\chi^2 =$

(v) **Critical region:**

(vi) **Conclusion:** D.Y.S.

VARIABLE:

A characteristic which can be measured numerically OR a quantitative characteristic is called as variable. *e.g. Height, Age, Weight, Number of students etc.*

ATTRIBUTES:

A characteristic which cannot be measured numerically but can be described by its presence or absence OR a qualitative characteristic is called as Attribute. *e.g. Gender, Habits, Eye color, Marital status etc.*

CONTINGENCY TABLE:

A table consisting of two attributes *A* and *B* in which attribute *A* is divided into *r* rows and attribute *B* is divided into *c* columns is called as *r* x *c* contingency Table.

Attributes		B				Total
		<i>B</i> ₁	<i>B</i> ₂		<i>B</i> _c	
<i>A</i>	<i>A</i> ₁	(<i>A</i> ₁ <i>B</i> ₁)	(<i>A</i> ₁ <i>B</i> ₂)		(<i>A</i> ₁ <i>B</i> _c)	(<i>A</i> ₁)
	<i>A</i> ₂	(<i>A</i> ₂ <i>B</i> ₁)	(<i>A</i> ₂ <i>B</i> ₂)		(<i>A</i> ₂ <i>B</i> _c)	(<i>A</i> ₂)
	⋮	⋮	⋮	⋮	⋮	
	<i>A</i> _r	(<i>A</i> _r <i>B</i> ₁)	(<i>A</i> _r <i>B</i> ₂)		(<i>A</i> _r <i>B</i> _c)	(<i>A</i> _r)
Total		(<i>B</i> ₁)	(<i>B</i> ₂)		(<i>B</i> _c)	<i>n</i>

- Note:** (i) The relationship between two variables is called correlation.
(ii) The relationship between two Attributes is called Association.

CHI-SQUARE TEST OF INDEPENDENCE IN *r* x *c* CONTINGENCY**TABLE:**

- (i) *H*₀: Two Attributes are Independent

*H*₁: Two Attributes are Associated

- (ii) **Level of Significance:**

- (iii) **Test Statistic:**

$$\chi^2 = \sum \left[\frac{(f_o - f_e)^2}{f_e} \right]$$

Where f_o = Observed frequency; f_e = Expected or theoretical frequency

- (iv) **Calculation:**

- (v) **Critical region:**

$$\text{If } \chi^2 \geq \chi^2_{(\alpha, d, f.)}$$

$$d.f. = (r-1)(c-1)$$

Then we shall reject H_0

(vi) **Conclusion:**

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Coefficient of Contingency For $r \times c$ Contingency Table OR Karl Pearson Coefficient of Contingency For $r \times c$ Contingency Table:

$$C = \sqrt{\frac{\chi^2}{n + \chi^2}}$$

It must be noted that $0 \leq C \leq \sqrt{\frac{k-1}{k}}$ where k is smaller from number of rows and columns i.e. $k = \min(r, c)$.

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Example# 6: Four hundred and ninety two candidates for scientific posts gave particulars of their university degrees and their hobbies. The degrees were in either maths, chemistry or physics and hobbies could be classified roughly as music, crafts work, reading or drama. The data are presented concisely in the following contingency table.

Categories	Maths	Chemistry	Physics
Music	24	83	17
Crafts work	11	62	28
Reading	32	121	34
Drama	10	26	44

Discuss the association between the two criteria of classification i.e. the degrees and the hobbies. Also find the Coefficient of Contingency C and **maximum value of C** and interpret the result?

Solution:

(i) H_0 : Two Attributes are Independent

H_1 : Two Attributes are Associated

(ii) **Level of Significance:**

$$\alpha = 0.05$$

(iii) **Test Statistics:**

$$\chi^2 = \sum \left[\frac{(f_o - f_e)^2}{f_e} \right]$$

(iv) Calculation:

Categories	Maths	Chemistry	Physics	Total
Music	24	83	17	124
Crafts work	11	62	28	101
Reading	32	121	34	187
Drama	10	26	44	80
Total	77	292	123	$n = 492$

Expected frequencies are;

Hobbies	Degrees			Total
	Mathematics	Chemistry	Physics	
Music	$\frac{124 \times 77}{492} = 19.4$	$\frac{124 \times 292}{492} = 73.6$	$\frac{124 \times 123}{492} = 31.0$	124
Crafts work	$\frac{101 \times 77}{492} = 15.8$	$\frac{101 \times 292}{492} = 59.9$	$\frac{101 \times 123}{492} = 25.3$	101
Reading	$\frac{187 \times 77}{492} = 29.3$	$\frac{187 \times 292}{492} = 111.0$	$\frac{187 \times 123}{492} = 46.7$	187
Drama	$\frac{80 \times 77}{492} = 12.5$	$\frac{80 \times 292}{492} = 47.5$	$\frac{80 \times 123}{492} = 20.0$	80
Total	77	292	123	$n = 492$

f_0	f_e	$(f_0 - f_e)$	$(f_0 - f_e)^2$	$\frac{(f_0 - f_e)^2}{f_e}$
24	19.4	4.6	21.16	1.09
83	73.6	9.4	88.36	1.20
17	31.0	-14	196.0	6.32
11	15.8	-4.8	23.04	1.46
62	59.9	2.1	4.41	0.07
28	25.3	2.7	7.29	0.29
32	29.3	2.7	7.29	0.25
121	111.0	10	100.0	0.90
34	46.7	-12.7	161.29	3.45
10	12.5	-2.5	6.25	0.50
26	47.5	-21.5	462.25	9.73
44	20.0	24.0	576.00	28.8
$\sum f_0 = 492$	$\sum f_e = 492$			$\chi^2 = 54.06$

$$\chi^2 = \sum \left[\frac{(f_0 - f_e)^2}{f_e} \right] = 54.06$$

(v) Critical region:

$$d.f. = (r-1)(c-1) = (4-1)(3-1) = 6$$

$$\text{If } \chi^2 \geq \chi^2_{(\alpha, d.f.)} = \chi^2_{(0.05, 6)} = 12.59$$

Then we shall reject H_0

(vi) Conclusion:

Since calculated value falls in the rejection region, therefore the null hypothesis is rejected. It is therefore concluded that there is association between the two criteria of classification *i.e.* the degrees and hobbies.

Coefficient of Contingency is

$$C = \sqrt{\frac{\chi^2}{n + \chi^2}} = \sqrt{\frac{54.06}{492 + 54.06}} = 0.31$$

$$\text{Maximum value of } C = \sqrt{\frac{k-1}{k}} = \sqrt{\frac{3-1}{3}} = 0.81 \quad \text{where } k = \min(r, c) = \min(4, 3) = 3$$

Interpretation:

$C = 0.31$ indicates that there is low association between two criteria of classification.

Example# 7: The following table shows the number of recruits taking (i) a preliminary and (ii) a final test in car driving. Use the χ^2 test to discuss whether there is any association between the results of preliminary and those of the final test?

Categories		Preliminary		Total
		Pass	Fail	
Final	Pass	605	135	740
	Fail	195	65	260
Total		800	200	1000

Example# 8: Test the association between injection against typhoid and exemption from attack from the following contingency table.

Classes	Attacked	Not-Attacked	Total
Inoculated	528	25	553
Not-inoculated	790	175	965
Total	1318	200	1518

Example# 9: The following table gives the census data of orchards. Test the hypothesis that the two variables of classification are independent.

Classes	Shaded	Un-shaded	Total
High Yields	350	205	555
Low Yields	250	195	445
Total	600	400	1000

Example# 10: Find χ^2 and test whether the two attributes are independent. Let $\alpha = 0.05$

Attributes	A_1	A_2	A_3	Total
B_1	215	325	60	600
B_2	135	175	90	400
Total	350	500	150	1000

Example# 11: A thousand households are taken at random and divided into three groups A, B and C according to the total monthly income. The following table shows the numbers in each group having a colour television receiver, a black and white receiver or no television at all.

	A	B	C
Colour T.V.	56	51	93
Black & White T.V.	118	207	375
None	26	42	32

Test the hypothesis that there is no association between total income and television ownership.

D.Y.S.

SHORTCUT FORMULA FOR THE COMPUTATION OF χ^2 IN 2 x 2 CONTINGENCY TABLE:

Attributes	B ₁	B ₂	Total
A ₁	a	b	a + b
A ₂	c	d	c + d
Total	a + c	b + d	a + b + c + d = n

$$\chi^2 = \frac{(ad - bc)^2 n}{(a + c)(b + d)(a + b)(c + d)}$$

YATE'S CORRECTION FOR CONTINUITY:

In applying χ^2 approximation, we are required to combine the smaller frequencies (less than 5) with larger ones. But in case of two classes, we cannot pool the smaller frequency into the larger one. For such a situation, Frank Yates in 1934 showed the χ^2 approximation is markedly improved by the following formula;

$$\chi^2 = \sum_{i=1}^2 \frac{\left(\left| f_o - f_e \right| - \frac{1}{2} \right)^2}{f_e}$$

OR

$$\chi^2 = \frac{\left(\left| ad - bc \right| - \frac{n}{2} \right)^2 n}{(a + c)(b + d)(a + b)(c + d)}$$

Example# 12: Given the following contingency table;

Categories	Immunized	Not immunized	Total
Infected	15	95	110
Not infected	50	160	210
Total	65	255	320

Does it seem that treatment gave any protection against infection? What is the difference in the significance of the result of the χ^2 test according as Yate's correction is or is not applied?

Solution:

(i) H_0 : Two Attributes are Independent

H_1 : Two Attributes are Associated

(ii) **Level of Significance:**

$$\alpha = 0.05$$

(iii) **Test Statistic:**

Without using Yate's Correction

$$\chi^2 = \frac{(ad - bc)^2 n}{(a + c)(b + d)(a + b)(c + d)}$$

Using Yate's Correction

$$\chi^2 = \frac{\left(|ad - bc| - \frac{n}{2} \right)^2 n}{(a + c)(b + d)(a + b)(c + d)}$$

(iv) **Calculation:**

Categories	Immunized	Not immunized	Total
Infected	$a = 15$	$b = 95$	$a + b = 110$
Not infected	$c = 65 - 15 = 50$	$d = 255 - 95 = 160$	$c + d = 210$
Total	$a + c = 320 - 255 = 65$	$b + d = 255$	$n = 320$

Without using Yate's Correction

$$\chi^2 = \frac{(15 \times 160 - 50 \times 95)^2 320}{(65)(255)(110)(210)} = 4.62$$

Using Yate's Correction

$$\chi^2 = \frac{\left(|(15 \times 160) - (50 \times 95)| - \frac{320}{2} \right)^2 320}{(65)(255)(110)(210)} = 4.01$$

(v) **Critical region:**

$$\text{If } \chi^2 \geq \chi^2_{(\alpha, d.f.)} = \chi^2_{(0.05, 1)} = 3.84$$

$$d.f. = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$$

Then we shall reject H_0

(vi) **Conclusion:**

Since calculated value falls in the rejection region in both cases, therefore the null hypothesis is rejected. It is therefore concluded that there is association between the two Attributes.

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